WaterQuest Solutions

STOP LEAKING OIL INTO THE FUTURE.

R2 / MODULAR REVERSE OSMOSIS

Fresh water. Built to grow.

Modular desalination packages

SEAWATER / BRACKISH WATER / PARALLEL OPERATION



Illustrative RO installation; not a WaterQuest project reference.

Start with the water you need. Build a practical path to more.

Modular RO concepts for communities, remote facilities and industry.

ONE MODULAR APPROACH. TWO FEEDWATER DUTIES.

Seawater or brackish water. Begin with the source.

The same production target can require a very different pump, membrane and treatment arrangement. Match the package to the actual feedwater and the quality you need.

SWRO

Seawater

For coastal communities, marine facilities and industrial sites using a saline intake. Review seasonal salinity, temperature, intake quality and concentrate management.

REFERENCE DUTY

55 bar / 798 psi

38% membrane recovery

BWRO

Brackish water

For saline groundwater and other lower-salinity supplies. Review scaling potential, fouling constituents and the treatment required to meet the product-water target.

REFERENCE DUTY

20 bar / 290 psi

75% membrane recovery

Reference duties are illustrative assumptions from the supplied overview, not guaranteed operating points. Final pressure and recovery come from the feed analysis and membrane projection.

DESIGN THE COMPLETE WATER PATHWAY

SOURCE Analyse the feed → **CONDITION** Prepare the water → **SEPARATE** Pump + membranes → **DELIVER** Quality + storage

SIX PLANNING TIERS

Start with daily demand.

Compare seawater and brackish-water production targets, then establish operating hours, storage and the capacity required during maintenance.

PLANNING TIER	SEAWATER / SW	BRACKISH / BW
Micro Remote and mobile sites	**6 m³/day** 1,585 US gal/day	**13 m³/day** 3,434 US gal/day
Small community Camps and small communities	**20 m³/day** 5,283 US gal/day	**46 m³/day** 12,152 US gal/day
Light municipal Small coastal facilities	**36 m³/day** 9,510 US gal/day	**82 m³/day** 21,662 US gal/day
Community municipal Utilities and aquaculture	**79 m³/day** 20,870 US gal/day	**178 m³/day** 47,023 US gal/day
Industrial Regional and industrial users	**235 m³/day** 62,080 US gal/day	**528 m³/day** 139,483 US gal/day
Utility scale Larger plants and expansion	**511 m³/day** 134,992 US gal/day	**1,147 m³/day** 303,005 US gal/day

Indicative production targets, not confirmed package ratings. Values are rounded from the supplied six-tier overview. Net delivery depends on feedwater, membrane selection, recovery, running hours and plant losses. Pump size, shaft speed and feed flow must be reconciled before a package is quoted.

Need more water? Add parallel trains around a defined expansion plan.

CHOOSE THE PUMPING ARRANGEMENT

Keep energy in the process.

Compare conventional feed pumping with integrated brine energy recovery at the same water output and feed conditions.

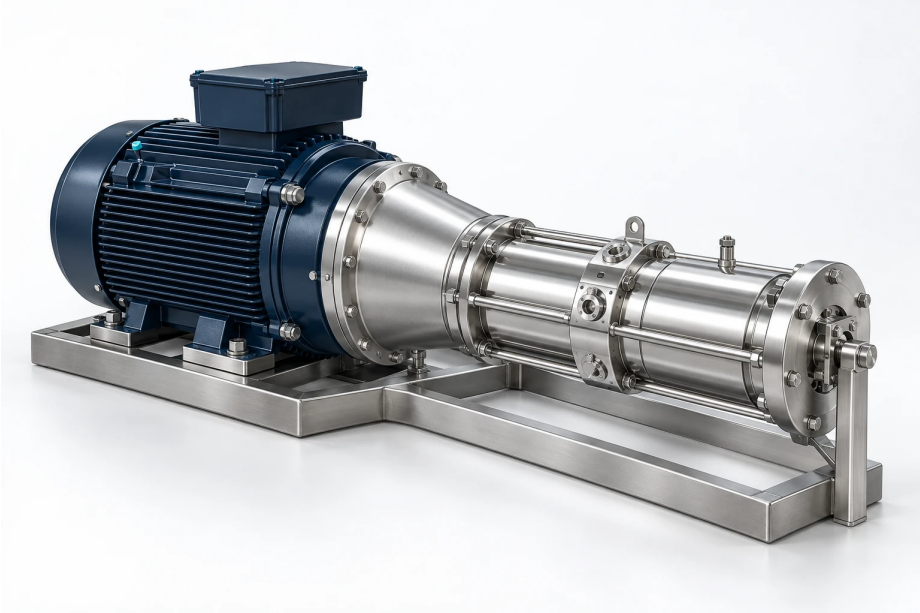


Illustration of a Power Centre assembly. Supplied configuration varies by duty.

Standard pump package

A high-pressure water pump supplies the RO membrane train. Select the drive and any separate energy-recovery equipment around the required flow, pressure and operating schedule.

Integrated Power Centre

Feed pumping and brine energy recovery work together. Recovering pressure energy from the concentrate stream can reduce the drive power needed for the selected RO duty.

COMPARE THE WHOLE PLANT

Specific energy (kWh/m³) = electrical input (kW) / net product water (m³/h).
Include pretreatment, feed boosting and other auxiliary loads in the comparison. Confirm savings at the selected duty; motor nameplate power is not measured energy use.

PARALLEL OPERATION

Run what you need. Keep room to grow.

Parallel RO trains divide production into serviceable modules. Stage the running units to meet demand, isolate a train for maintenance and add capacity when the site is ready.

A MODULAR PLANT ARCHITECTURE



Concentrate from each train follows its specified recovery and disposal route.

01 / MATCH DEMAND

Bring trains on or offline as demand changes. Variable-speed control can trim output within the selected pump and membrane operating limits.

02 / SERVICE ONE TRAIN

Provide isolation, safe depressurisation, flushing and access. Remaining output depends on the capacity still available and any usable storage.

03 / COORDINATE THE DRIVES

A trim-drive arrangement may use one variable-speed unit with other units at fixed speed. Confirm turndown, starting, pressure control and each train's stable operating range.

04 / EXPAND IN STAGES

Reserve manifold, pretreatment, electrical and site capacity for future modules. Expansion works best when these shared services are planned at the outset.

PARALLEL OPERATION / WORKED EXAMPLE

Plan for the train that is not running.

Illustrative seawater case: three equal trains, each planned at 511 m³/day (135,000 US gal/day), operating over 24 hours. Figures are rounded planning values, not a selected pump rating.

ALL TRAINS RUNNING

1 2 3

1,533 m³/day

405,000 US gal/day

Combined daily target

ONE TRAIN ISOLATED

1 2 3

1,022 m³/day

270,000 US gal/day

Remaining daily target

ONE TRAIN RUNNING

1 2 3

511 m³/day

135,000 US gal/day

Lower-demand operating stage

A 1,500 m³/day (396,000 US gal/day) requirement is not maintained by two of these trains. If that output must continue with one train unavailable, assess an additional train, a different train size or a defined storage strategy.

Shared intake, pretreatment, electrical supply, headers and controls remain potential common failure points. Define the outage that the redundancy covers and demonstrate the required net delivery during commissioning.

FROM PACKAGE CONCEPT TO OPERATING PLANT

Build around the site. Make service practical.

The pump is one part of the installation. Define the water balance, the treatment scope and the interfaces so the complete plant can deliver the agreed result.

01 Feedwater and pretreatment

Start with seasonal chemistry, salinity, temperature and fouling potential. Establish intake, filtration and inlet conditions for the selected pump and membranes.

02 Materials and membranes

Review water chemistry, concentrate and cleaning agents against wetted materials and seals. Finalise membrane elements, pressure vessels and recovery with the membrane supplier.

03 Integration and protection

Define the motor and controls, pressure protection, instrumentation, pipework, lifting access and footprint. Confirm any electrical or hazardous-area approvals for the exact configuration.

04 Service and availability

Include safe isolation, flushing, preservation, spares and restart procedures. Set inspection and service intervals using the current equipment instructions and actual operating duty.

Close the water balance.

Product water + concentrate + other plant losses must reconcile with the feed supply.

YOUR NEXT WATER PROJECT

Bring us the source. Tell us the demand.

We'll help define the package.

WaterQuest Solutions provides Canadian support for water-hydraulic components and RO package selection. Start with the water you have, the water you need and the way your site operates.



Illustrative membrane-train arrangement.

Send a feedwater analysis, daily and peak demand, operating hours, product-water quality target and any required maintenance reserve. A site sketch and utility details help define the next step.

Discuss your desalination project

bruce@waterquestsolutions.com
+1 (604) 902-1902

waterquestsolutions.com
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